

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Centre of Excellence in Artificial Intelligence

2. **Subject Name:** Reinforcement Learning **L-T-P:** 3-1-0 **Credits:** 4

3. **Pre-requisites:** Probability and Statistics

4. **Syllabus and reference books:**

Syllabus:

Module 1: Introduction

- Motivating examples and simulation environments (Open AI Gym and safe control Gym)
- Fundamentals without noise

Module 2: Fundamentals

- Markov chains and state-space models
- Markov chains and state-space models
- Markov Decision Processes
- Multi-armed Bandits, Upper Confidence Bounds and Thompson Sampling

Module 3: Stochastic Optimal Control

- Finite horizon dynamic programming
- Infinite horizon MDP
- Existence of the optimal policy

Module 4: Tabular Solution Methods

- Value and policy iteration algorithms
- Temporal-difference prediction and optimality
- SARSA: On-policy TD control
- Q Learning: Off-policy TD control

Module 5: Approximate Solution Methods

- On-policy prediction and control
- Off-policy function approximation
- Eligibility Traces

Module 6: Case Studies

- Deep RL
- Safe RL
- Project presentations

Reference Books:

- 1) **Reinforcement Learning: An introduction**, R.S. Sutton and A.G. Barto, 2nd Edition, MIT Press, 2018.

- 2) **Control Systems and Reinforcement Learning**, Sean Meyn, Cambridge University Press, 2022.
- 3) **Markov Decision Processes: Discrete Stochastic Dynamic Programming**, M.L. Puterman, 1st Edition, Wiley, 2005.
- 4) **Dynamic Programming and Optimal Control**, D.P. Bertsekas, Vol. I, 4th Edition, Athena Scientific, 2017.
- 5) **Dynamic Programming and Optimal Control**, D.P. Bertsekas, Vol. II, 4th Edition, Athena Scientific, 2012.
- 6) **Bandit Algorithms**, T. Lattimore and C. Szepesvári, 1st Edition, Cambridge University Press, 2020.
- 7) **Lecture Notes on Markov Decision Process** by Lodewijk Kallenberg.
- 8) **Lecture notes on Foundations of Intelligent and Learning Agents** by Shivaram Kalyanakrishnan.
- 9) **Lecture notes on Control of Stochastic Systems** by R. Srikant.
- 10) **Stochastic Systems: Estimation, Identification and Adaptive Control**, P. R. Kumar and P. Varaiya, SIAM, 2015.

5. Lecture-wise break-up:

S. N.	Module	Topic	No. of hours
1	Introduction	Motivating examples and Simulation environments (Open AI Gym and safe control Gym)	2 hours
2		Fundamentals without noise	3 hours
3	Fundamentals	Markov chains and state-space models	3 hours
4		Markov Decision Processes	3 hours
5		Multi-armed Bandits, Upper Confidence Bounds and Thompson Sampling	3 hours
6	Stochastic Optimal Control	Finite horizon dynamic programming	2 hours
7		Infinite horizon MDP	2 hours
8		Existence of the optimal policy	2 hours
9	Tabular Solution Methods	Value and policy iteration algorithms	3 hours
10		Temporal-difference prediction and optimality	3 hours

11		SARSA: On-policy TD control	3 hours
12		Q Learning: Off-policy TD control	3 hours
13	Approximate Solution Methods	On-policy prediction and control	3 hours
14		Off-policy function approximation	3 hours
15		Eligibility Traces	2 hours
16	Case Studies	Deep RL	3 hours
17		Safe RL	3 hours
18		Project presentations	2 hours
Total number of hours			48 hours